

Actividades Unidad 3 Métodos Numéricos

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Ejercicios a libreta

3.1 Eliminación Gaussiana

E1

$$2x + y = 5$$

$$x + 3y = 6$$

$$\begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 6 \end{bmatrix}$$

$$2x + 6y = 12$$

$$(2x + 6y) - (2x + y) = 12 - 5$$

$$5y = 7$$

$$y = \frac{7}{5} = 1.4$$

$$2x + 1.4 = 5$$

$$2x = 3.6$$

$$x = 1.8$$

$$x = 1.8, y = 1.4$$

E2

$$\begin{aligned}x + y + z &= 6 \\2x + 3y + z &= 14 \\x + 2y + 3z &= 14\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 14 \\ 1 & 2 & 3 & 14 \end{array} \right] \quad \begin{aligned}F_2 &= F_2 - 2F_1 \\F_3 &= F_3 - F_1\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & 2 & 8 \end{array} \right]$$

$$F_3 = F_3 - F_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 3 & 6 \end{array} \right]$$

$$3z = 6$$

$$z = 2$$

$$y - z = 2$$

$$y - 2 = 2$$

$$y = 4$$

$$x + y + z = 6$$

$$x + 4 + 2 = 6$$

$$x = 0$$

$$x = 0, y = 4, z = 2$$

E3

$$3x - y + z = 4$$

$$x + 2y - z = 1$$

$$2x + y + 2z = 5$$

$$\begin{bmatrix} 3 & -1 & 1 & 4 \\ 1 & 2 & -1 & 1 \\ 2 & 1 & 2 & 5 \end{bmatrix}$$

$$F_2 = F_2 - 3F_1$$

$$F_3 = F_3 - 2F_1$$

$$\begin{bmatrix} 3 & -1 & 1 & 4 \\ 0 & -7 & 4 & 1 \\ 0 & -3 & 0 & 3 \end{bmatrix}$$

So eliminating y.

$$F_3 = 7F_3 - 3F_2$$

$$16z = 18$$

$$z = \frac{18}{16} = \frac{9}{8} = 1.125$$

$$-7y + 4z = 1$$

$$-7y + 4(1.125) = 1$$

$$-7y + 4.5 = 1$$

$$-7y = -3.5$$

$$y = 0.5$$

$$x + 2y - z = 1$$

$$x + 2(0.5) - 1.125 = 1$$

$$x + 1 - 1.125 = 1$$

$$x = 1.125$$

$$x = 1.125, y = 0.5, z = 1.125$$

E 4

$$4x + 3y = 10$$

$$2x + 5y = 12$$

$$\begin{bmatrix} 4 & 3 & | & 10 \\ 2 & 5 & | & 12 \end{bmatrix}$$

$$F_2 = 2F_2 - F_1$$

$$\begin{bmatrix} 4 & 3 & | & 10 \\ 0 & 7 & | & 14 \end{bmatrix}$$

$$7y = 14$$

$$y = 2$$

$$4x + 3(2) = 10$$

$$4x + 6 = 10$$

$$4x = 4$$

$$x = 1$$

$$\underline{x = 1, y = 2}$$

3.2 Gauss-Jordan

E5

$$2x + y - z = 8$$

$$-3x - y + 2z = -77$$

$$-2x + y + 2z = -3$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -77 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} -3 & -1 & 2 & -77 \\ 2 & 1 & -1 & 8 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

Por lo tanto:

$$x=2, y=3, z=-1$$

Ej 6 a) Sup x (original) solution and x is feasible (x) on $x \geq 0$ and $x \leq 1$

Encontrar la Inversa

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & | & 1 & 0 \\ 1 & 3 & | & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & | & 0 & 1 \\ 2 & 1 & | & 1 & 0 \end{bmatrix}$$

$$F_2 = F_2 - 2F_1$$

$$\begin{bmatrix} 1 & 3 & | & 0 & 1 \\ 0 & -5 & | & 1 & -2 \end{bmatrix}$$

$$F_2 = \frac{F_2}{-5}$$

$$\begin{bmatrix} 1 & 3 & | & 0 & 1 \\ 0 & 1 & | & -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$$